

ICS 4111: Embedded Systems & IoT

Semester Project: Deliverable 1

Objective: Identify requirements that support individual flower growth and develop schematic designs of embedded devices

Instructions

1. In your groups for the semester project, research on the environmental requirements that support proper growth of the flower as assigned to your group. Briefly describe and document the following characteristics for the growth of the flower:

- a. Optimal temperature range
- b. Optimal relative humidity range
- c. Recommended soil type
- d. Optimal soil moisture content
- e. Optimal soil pH range
- f. Suitable number of hours to sunlight exposure.

Have these measurements in a table that the team will reference to later in the project.

2. Identify and list **all** suitable hardware components that you will need as a team to develop an embedded device for monitoring the environmental metrics listed in number 1. Part of the environmental measurements includes amount of LPG (methane/butane/propane).

*Note, that **all hardware components** include tools used for prototyping e.g. breadboards and jumper wires.*

3. Retrieve datasheets for these components and include their links to your project document. If datasheets are unavailable, you can include webpages describing the product:
 - a. 1.3" White IIC 128X64 OLED LCD
 - b. ESP32S DevKIT WIFI + BLE Module (30Pin)
 - c. DHT22 AM2302 Temperature and Humidity sensor

- d. MQ-5 LPG, natural gas, coal gas Sensor
 - e. 5V 1-Channel Low Level Trigger Relay Module
4. Develop schematic diagrams of the following architectures based on the listed components in number 3, each listed item represents one design.
- a. 1 ESP32S connected to 1 MQ-5, 1 DHT22 and 1 LCD
 - b. 1 ESP32S connected to 1 MQ-5 interfaced directly with another ESP32S connected to 1 DHT22
 - c. 1 ESP32S connected to 1 DHT22 connected to 1 relay which is connected to another ESP32S connected to 1 MQ-5

Note: the expected level of detail in your circuit diagrams should be similar to figure 7 in [this research paper](#) or the example given on e-learning. Include appropriate electric components such as resistors, capacitors and voltage dividers in your circuits.

Once done, generate a Markdown document (.md) in your GitHub repo and include all your content for the deliverable in the document. You can refer to formatting guidelines for Markdown docs [here](#). Submit the project repo link to e-learning before the due date. Remember to include evidence of groupwork in your submission.